RealSense™ application. It should be based on one of the following categories:

- Gaming & Play
- Learning & Edutainment
- Interact Naturally
- Immersion, Collaboration and Creation
- Open Innovation

The estimated timeline for this challenge is as follows. The Ideation phase will begin in May 2014 with the Development phase set to kick off in July 2014. Subsequently, the winners are expected to be announced near the end of 2014. For more information and alerts related to this challenge, go to: *https://software.intel.com/sites/campaigns/realsensechallenge/*

PERCEPTUAL COMPUTING CHALLENGE WINNERS SHOWCASE

ARPEDIA

As part of its Perceptual Computing campaign last year, Intel had announced the Perceptual Computing Challenge. The winners of the Phase 2 of this challenge in the 'Creative User Experience' category were announced some time back and their applications were demoed at CES 2014. Six team members from the Beijing University of Technology developed an application called 'ARPedia' using the Augmented Reality features. The application's 3D models were built using Maya 3D while Unity 3D was used to complete the application logic and render scenes. The application uses hand gestures, voice and touch commands and augmented reality integration to develop an encyclopedia similar to Wikipedia that responds to these newer modes of interaction.

For example, the demo showed that users could pick up a Tyrannosaurus Rex image and place it at different parts of the screen to know about different types of details related to this dinosaur. The application has very good interactions and builds an immersive experience by evoking involvement of the user with the virtual environment.

For a case study related to this application, you can visit the page here: *https://software.intel.com/en-us/articles/arpedia-case-study*

HEAD OF THE ORDER

Head of the Order is a spell casting game developed by Jacob and Melissa Pennock from Unicorn Forest Games. Last year, it took the top prize in the Intel Perceptual Computing Challenge. The game succeeds since it requires zero physical contact. Opponent players can play it in beautiful indoor and outdoor environments and create spells involving fire, ice and shields using hand gestures.

Jacob developed his own gesture recognition system to improve the accuracy of the established algorithms and successfully used his HyperGlyph library in multiple games. The hand rendering system also allows the game to represent the image of the hands on the screen. There are a number of other innovative features in the game such as 'Gesture Sequencing' to perform unique actions once gestures are performed in a particular sequence. For more information related to this game, visit the case study at: *https://software.intel.com/en-us/articles/head-of-the-order-case-study*

In conclusion, the Intel® RealSense™ technology provides an innovative framework spanning hardware as well as software to take the lead in developing innovative ways of interacting with computers. Going forward, we can expect to see more and more interfaces in desktop as well as mobile platforms which will surpass the traditional methods to include more natural ways of interacting with software. This will not only make things easier, but will also allow us to capture more data such as heart pulses and let the software react to minor cues such as emotions or facial expressions. Just like the new age touchscreen phones, which replaced the hardware keyboard in earlier phones, RealSense™ hardware is expected to revolutionize the way we see input devices today. With the current push towards making things smaller and more portable, it has already become a challenge to figure out a way to integrate all this hardware in every form factor possible but Intel is working hard to ensure that we see this technology everywhere around us. ■